

STAT 3003: Survey Methods **- Introduction**

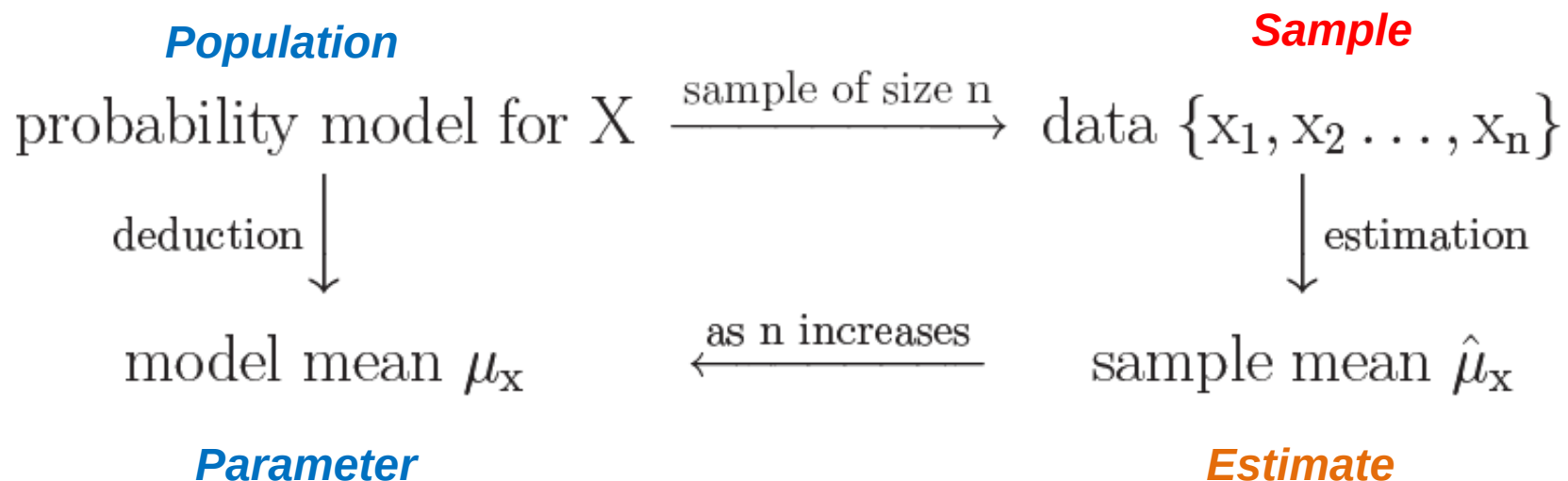
Instructor: Professor FAN, Xiaodan (Dan, 樊曉丹)

What is Survey

- A statistical survey is any structured inquiry designed to obtain data, which may be qualitative or quantitative.
- The main goal of a survey is to collect data that is representative of the population, allowing researchers to make informed decisions or draw conclusions.
- Survey sampling describes the process of selecting a sample of elements from a target population to conduct a survey.
- Survey Methods teach how to sample elements from the population, how to construct questionnaire, how to improve the number and accuracy of responses to surveys, and how to analyse the resulting data.

Role in Statistics

- A **variable** is a characteristic measured on individuals/elements drawn from a population under study.
- **Data** are measurements of one or more variables made on a collection of individuals.
- Variable/Data Type: discrete, continuous, categorical



Role in Common Path of Statistical Studies

- 1. Scientific Goal

e.g. find disease gene, compare efficacy of 2 drugs, ...

- 2. Study Design

e.g. observation, experiment, cohort study, case-control, ...

- 3. Data Collection

e.g. what variables including confounders to collect, how to collect, ...

- 4. Data Describing

e.g. histograms, box plots, average, ...

- 5. Statistical Inference

e.g. estimation, hypothesis testing, prediction, ...

- 6. Interpretation

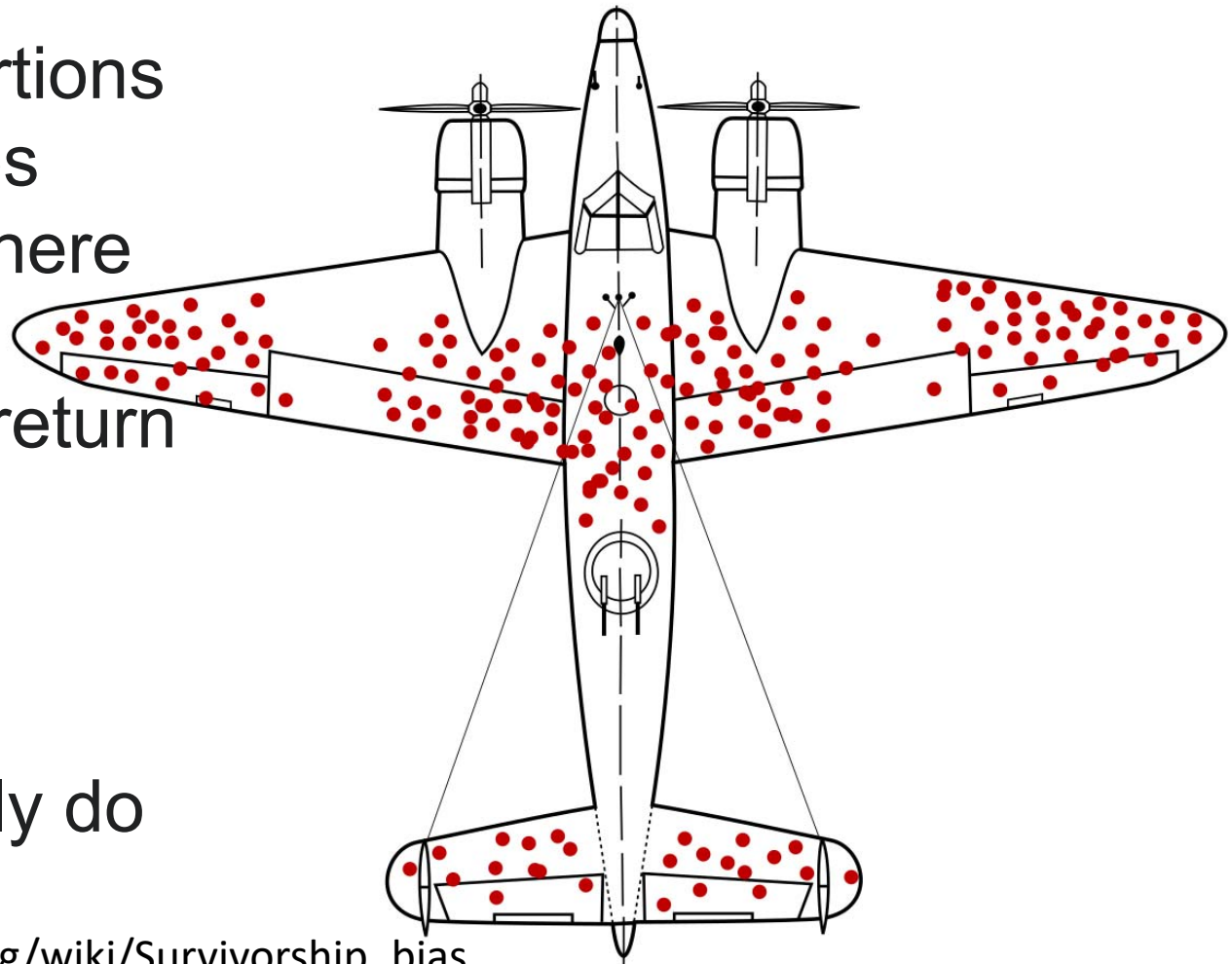
Motivating Example 1:

IMPORTANT to **understand your data:**

Survival Bias - a form of selection bias

The damaged portions
of returning planes
show locations where
they can sustain
damage and still return
home.

Those hit in other
places presumably do
not survive.



Ref: https://en.wikipedia.org/wiki/Survivorship_bias

Motivating Example 2:

IMPORTANT to **understand your data:**
selection bias and nonresponse

Example 1.1 from Lohr book: **“The Survey That Killed a Magazine”** :

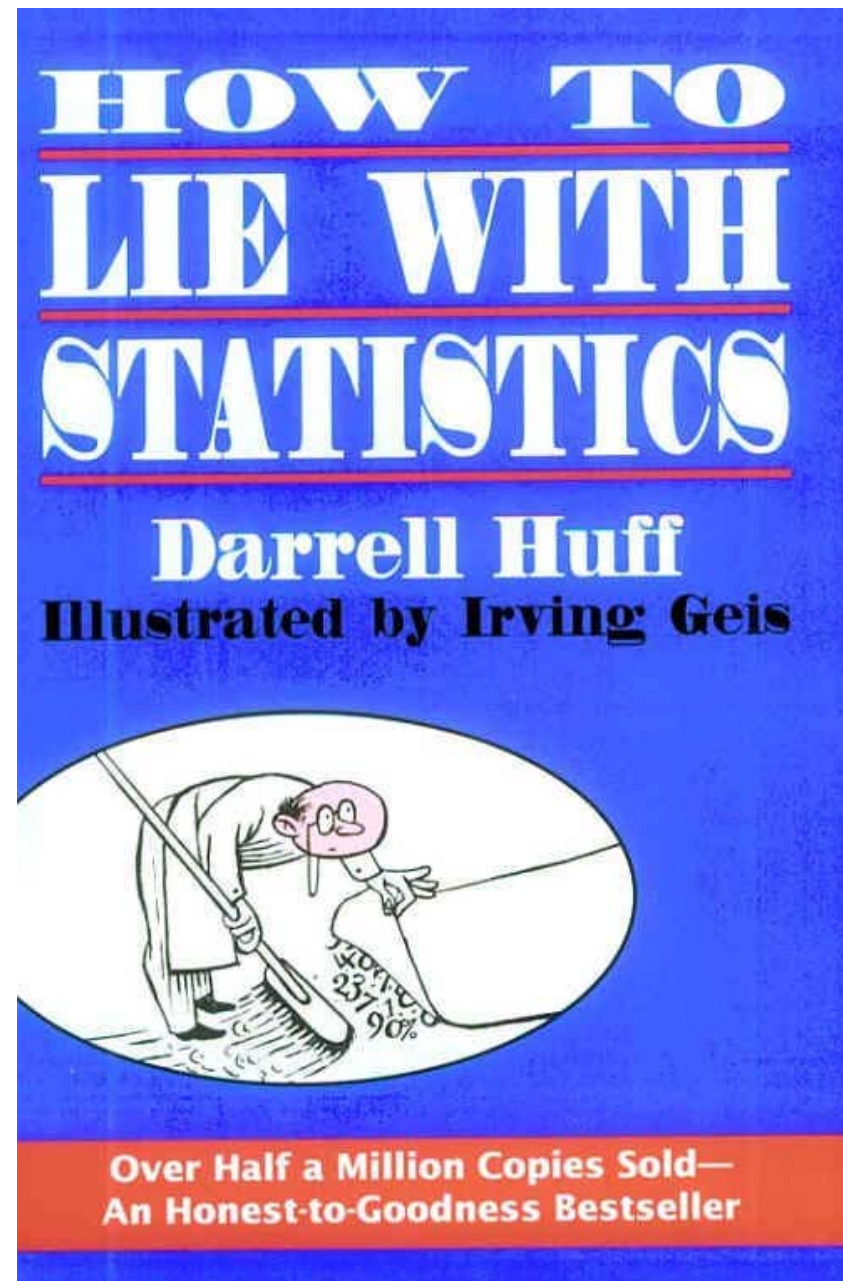
The Literary Digest, a weekly news magazine founded in 1890, did a presidential election poll in 1936 and predicted that Republican Alf Landon would win, but the later election showed that Democrat Franklin Roosevelt won with a big margin.

*names of persons to be polled were compiled from sources such as telephone directories and automobile registration lists => **bias** to rich people

* Ten million questionnaires were mailed out, and more than 2.3 million were returned => **nonresponse**, and persons supporting Landon were much **more likely to response** to the survey

Quotations

- There are three kinds of lies: lies, damn lies, and statistics.
--Mark Twain => Benjamin Disraeli
- It is easy to lie with statistics, but easier to lie without them.
--Frederick Mosteller



alert for statistics

- When statistics are not based on strictly accurate calculations, they mislead instead of guide. The mind easily lets itself be taken in by the **false appearance of exactitude** which statistics retain in their mistakes, and confidently adopts **errors clothed in the form of mathematical truth**.

—Alexis de Tocqueville, *Democracy in America*

Syllabus/Course Outline

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Logistics

- Homework: announce in class, available online, due before class, **submit through Blackboard**, R is recommended for programming
- Midterm policy: closed book, but allow 1 page note (A-4, double side)
- Final Exam policy: close-book, but allow 1 page note (A-4, double side)

Reminder:

- This course assumes basic statistics and probability background, and you will have to deal with quite many formulas.
- Life is short, so use it on useful and/or interesting things
- If you select it, devote to it. Don't waste your time
- Respect others' right to learn, as well as respect yourself

Background Checking

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Introduction Contents

- 1.1 The Problem
 - The goal of sampling, potential errors, example
- 1.2 Definitions of Population Parameters
 - μ, σ^2, S^2 (the corrected population variance),
 $Cov(X, Y), \rho$
- 1.3 Estimation
 - $\theta, \hat{\theta}$, sampling distribution, error of estimation,
margin of error, n , confidence interval (CI), bias, MSE
- 1.4 Survey Terminology
 - Element, population, sampling unit, frame, sample,
example

1.1 The Problem

- How to infer facts about a population from the information within a sample?
 - The average height of statistics students in Hong Kong
 - The proportion of statistics students in Hong Kong who wear glasses
 - The number of statistics students in Hong Kong who support the use of ChatGPT for homework

1.1 The Problem

- How to infer facts about a population from the **information** within a sample, *given a budget?*
- Each observation (a measurement, answers on a questionnaire) contains information
- But each observation also **costs** something (time and money)

1.1 The Problem

- Our resources are finite
- Too little information and we will inevitably make a poor estimate
- Too much information means we have wasted time/money
- What does the amount of information in a sample depend on?

1.1 The Problem

- The amount of information in a sample depends on the **sample size** and the **amount of variation** in the data
- In this course we will see many methods to control the amount of variation – these are called **sample survey designs**

1.1 The Problem

- Survey design and sample size determine the amount of information in the sample, assuming that each element in the sample is measured accurately
 - How much do you earn?
 - “Does it seem possible to you that the Holocaust never happened?” versus “Does it seem possible or does it seem impossible to you that the Nazi extermination of the Jews never happened?”
 - Have you ever copied your homework?

1.1 The Problem

- How much do you earn?
 - People tend to inflate their true wage to impress the person conducting the questionnaire
- Does it seem possible or does it seem impossible to you that the Nazi extermination of the Jews never happened?
 - The double negative is confusing. A clearer version of this question reduced the 'Yes' response from 22% to 1%
- Have you ever copied your homework?
 - This question is deeply embarrassing to answer

1.1 The Problem

- Problems like these can't be handled using statistical techniques which process the data after it has been collected
- To address these kinds of errors, the questionnaire/survey should be **planned carefully**
- We shall see methods to do this

1.1 The Problem

- We will want to **summarize information** about populations and samples
- We treat the results obtained in sample surveys as **random**, because we will **randomly select** members of the population to be in our sample
 - Why would we want to do this?
- Therefore different people using the same methods can obtain different results

1.1 The Problem

- Example – The McDonalds Survey
 - Our survey consists of one question: “How many times a month do you eat food from McDonalds?”
 - The population is everyone in this lecture theatre, now
 - Answer the question and write down your answer

1.1 The Problem

- Example – The McDonalds Survey
 - Our survey consists of one question: “How many times a month do you eat food from McDonalds?”
 - The population is everyone in this lecture theatre, now
 - Answer the question and write down your answer
 - I need 3 volunteers...

1.1 The Problem

- Example – The McDonalds Survey
 - Each volunteer is interested in how much McDonalds food this class consumes
 - But they do not have enough time or much to ask everyone
 - So they randomly select a sample of 3 each
 - And they collect their results...

1.1 The Problem

- Example – The McDonalds Survey
 - What can they do with their results?
 - With the information they have, how can they summarize their sample?
 - Can they infer anything about the population from their data?

1.1 The Problem

- The McDonalds Survey
 - They might average the responses they received
 - They might work out the standard deviation of the responses they received
 - Then each volunteer will have **estimates** of the population mean and standard deviation
 - Let's compare their estimates to these population **parameters**

1.1 The Problem

- Example – McDonalds Survey
 - Think about: how could the volunteers improve their surveys?

1.2 Definitions of Population Parameters

- We introduced population mean and population standard deviation and their sample versions, the sample mean and sample standard deviation
- Before we start creating survey designs, we need to remember some fundamental definitions and concepts from statistics

1.2 Definitions of Population Parameters

- **Population Mean**

- Let there be N people in the population. Let the variable of interest be Y (e.g. height in cm) and let person i have value u_i (e.g. the 1729th person has height 180 cm)
- Is the **average value** of all measurements in the population, i.e. $\frac{u_1 + u_2 + \dots + u_N}{N} := \mu$
 - We will often denote the population mean by μ
- The population mean in our McDonalds example is...

1.2 Definitions of Population Parameters

- **Finite Population Variance**

- The finite population variance is given by

- $$\frac{1}{N} \sum_{i=1}^N (u_i - \mu)^2 := \sigma^2$$

- We often denote the population variance by σ^2

- The **population standard deviation** (SD) simply the square root of the population variance

- $$\sqrt{\sum_{i=1}^N \frac{(u_i - \mu)^2}{N}} := \sigma$$

- The population standard deviation in our McDonalds example is...?

1.2 Definitions of Population Parameters

- In some textbooks, we may see finite population variance defined as

$\frac{1}{N-1} \sum_{i=1}^N (u_i - \mu)^2 := S^2$, the so-called “corrected population variance”

- Comparing this definition to our σ^2 , notice

$S^2 = \left(\frac{N}{N-1}\right) \sigma^2$, so as N become large, they are more or less the same

- Why bother with S^2 ?

Because the “ $\frac{1}{N-1}$ ” factor simplifies many formulas, as we’ll see later.

- I prefer σ^2 to S^2 as it better reflects the idea of variance as an expectation

$$\text{Var}(Y) = E[(Y - \mu)^2]$$

1.2 Definitions of Population Parameters

- It will be useful to have some measure of dependence between two random variables
- Some variables are clearly **independent**
 - Level of the Ulan Bator Stock Index and my shoe size
- But some obviously have some **dependence**
 - FTSE100 stock index and UK gilt prices
- **Covariance** and **Correlation** help us quantify dependence

1.2 Definitions of Population Parameters

- **Covariance**

- We have a finite population of size N . Each person in the population has two variables of interest, Y and X
- Let the population means of Y and X are μ_Y and μ_X respectively. Then the population covariance of X and Y is given by
 - $\frac{1}{N} \sum_{i=1}^N [(y_i - \mu_Y)(x_i - \mu_X)] := Cov(X, Y)$
- If X and Y are independent, then $Cov(X, Y)$ is zero
- The converse is not necessarily true
- It is tricky to decide whether a covariance is large or small because the scales of y and x are combined
- It would be better to have a dimensionless number to describe dependence...

1.2 Definitions of Population Parameters

- **Correlation**
 - Is that dimensionless quantity
 - The correlation between X and Y is just the covariance of X and Y divided by the standard deviations of X and Y
 - That is, $\frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y} := \rho$
 - This must lie between -1 and 1 and it has no units
 - We often denote it by ρ (rho)

1.2 Definitions of Population Parameters

- Population mean, variance and correlation are examples of parameters
 - **Parameters** are numerical descriptive measures of a population (mean, **total**, variance, **proportion**, **population size**, etc.)
 - Unless we have measured every member of the population, parameters will always be **unknown**

1.3 Estimation

- Much of this course will be spent on investigating how we can best **estimate** the parameters of a population from a sample of the population
 - We'll take the observations from our sample, plug them into a formula that gives our guess for the value of the parameter
 - That guess from our sample is called the **estimate**. The general formula is called the **estimator**.

1.3 Estimation

- Example
 - From our McDonald's surveys, what were the **parameters**, **estimates** and **estimators**?
- We denote a generic parameter by θ (theta)
- We denote an estimator of that parameter by putting a hat on it - $\hat{\theta}$
- Notice that, before we conduct the survey, $\hat{\theta}$ is a **random variable**
 - Because the elements in the sample were chosen **randomly**
- Thus $\hat{\theta}$ will have a probability distribution
 - This distribution is called the **sampling distribution**

1.3 Estimation

- Why do we care about the sampling distribution?
 - Because it is not enough to just present a single number as your best guess for the value of θ
 - This gives no idea about how good or accurate your guess is
- Your estimate for θ is very unlikely to be exactly correct and indeed, we wouldn't know if it was, because we'll never know the true value of θ
 - But it would be nice to know that the chances of it being wildly incorrect are quite small

1.3 Estimation

- That is, we would like our guess to be close to θ , most of the time
 - i.e. we would like $|\hat{\theta} - \theta|$ to be smaller than, say, some number d , most of the time
 - $|\hat{\theta} - \theta|$ is called the **error of estimation**
- We can write this in terms of a probability statement
 - We would like $\Pr(|\hat{\theta} - \theta| < d) = 1 - \alpha$, where α is a small number like 0.05 or 0.01 and d is called the **margin of error**
- The only random variable in the statement above is $\hat{\theta}$. In order to choose a sensible number d , we must have some idea **what the sampling distribution is**

1.3 Estimation

- The distribution of $\hat{\theta}$ will depend crucially on n , the size of the sample
- If n is small, the distribution of $\hat{\theta}$ is usually complicated and unknown
- If n is large, the distribution of $\hat{\theta}$ is much more regular and is well approximated by distributions we know well
- If n is large and close to N , the distribution of $\hat{\theta}$ tends to concentrate more and more around the true value of θ , but is not well approximated by distributions we know well

1.3 Estimation

- Imagine we are confident enough to say $\frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} \sim N(0,1)$, where $\sigma_{\hat{\theta}}$ is the standard deviation of $\hat{\theta}$
- Then, we may write

$$P\left(\frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} < z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$
$$\Rightarrow P\left(-z_{1-\frac{\alpha}{2}} < \frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} < z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

1.3 Estimation

- Carrying on, we have

$$\Rightarrow P\left(-\sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < \hat{\theta} - E[\hat{\theta}] < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(-\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < -E[\hat{\theta}] < -\hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < E[\hat{\theta}] < \hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

- And if $\hat{\theta}$ is unbiased, i.e. if $E[\hat{\theta}] = \theta$, then

$$P\left(-\sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < \hat{\theta} - \theta < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(|\hat{\theta} - \theta| < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha, \text{ meaning we have found}$$

our sensible number, d

1.3 Estimation

- The interval $\left(\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}, \hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right)$ is random
 - recall $\hat{\theta}$ is a random variable
 - If we plug in our observation of $\hat{\theta}$ from our sample, we have a **$100 \times (1 - \alpha)\%$ confidence interval for parameter θ**
- Having a **normal sampling distribution** is very handy. We will discuss conditions for its existence in the next lecture.
- Clearly we want our estimator $\hat{\theta}$ to have the property “ $E[\hat{\theta}] = \theta$ ”. What else do we want from our estimator?

1.3 Estimation

- We want our observation of $\hat{\theta}$ from our sample to be close to the parameter we're interested in
- This can happen if
 - The estimator generates estimates **are correct on average**
 - And these estimates are **not too spread out**
- Statistically, we want
$$E[\hat{\theta}] = \theta, \text{ as before}$$
$$E[(\hat{\theta} - \theta)^2] \text{ is small}$$

1.3 Estimation

- The quantity $E[\hat{\theta}] - \theta$ is called the **bias** of the estimator $\hat{\theta}$
 - We want our estimators to be **unbiased**, i.e. to have zero bias
- The quantity $E[(\hat{\theta} - \theta)^2]$ is called the **mean square error** (MSE) of $\hat{\theta}$
- There is a relationship between MSE, bias and variance

1.3 Estimation

- $$\begin{aligned} E \left[(\hat{\theta} - \theta)^2 \right] &= E \left[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2 \right] \\ &= E \left[(\hat{\theta} - E[\hat{\theta}])^2 \right] + 2(E[\hat{\theta}] - \theta)E \left[\hat{\theta} - E[\hat{\theta}] \right] \\ &\quad + (E[\hat{\theta}] - \theta)^2 \\ &\Rightarrow \textcolor{red}{MSE}(\hat{\theta}) = \textcolor{red}{Var}(\hat{\theta}) + \textcolor{red}{Bias}(\hat{\theta})^2 \end{aligned}$$
- Therefore, **if estimator $\hat{\theta}$ is unbiased** (most of our estimators will be unbiased), then its **MSE equals its variance**

1.3 Estimation

- Here are three useful words to describe an estimator
 - Unbiased if $E[\hat{\theta}] = \theta$
 - **Precise** if $Var(\hat{\theta})$ is small
 - **Accurate** if $E[\hat{\theta}] = \theta$ and $E[(\hat{\theta} - \theta)^2]$ is small
- Thus a precise estimator may not be accurate because it is biased.
- Precision considers how close estimates from different samples are to each other.
- Accuracy considers how close estimates are to the true value.

1.4 Survey Terminology

- We will go through some unavoidable technical terms which we will use repeatedly throughout the course
- **Element**
 - An element is an object on which a measurement is taken
 - E.g. a Hong Kong statistics student, an orange, anything

1.4 Survey Terminology

- **Population**
 - A collection of elements about which we want to make inferences
 - E.g. Hong Kong statistics students, oranges grown in California

1.4 Survey Terminology

- Sampling Units

- Non-overlapping collections of elements from the population that cover the entire population (i.e. they partition the population)
 - E.g. CUHK statistics students, HKU stats students, HKUST stats students... etc. (assuming that a student cannot study statistics in two university in Hong Kong at the same time)
 - Often sampling units and elements are the same

1.4 Survey Terminology

- **Frame**
 - A frame is a list of sampling units
 - E.g. Lists of registered CUHK statistics students, of HKU statistics students, etc.
 - ‘List’ can be loosely interpreted: any representation of the population that allows random selection of units can be a frame

1.4 Survey Terminology

- Sample
 - A collection of sampling units drawn from a single or from multiple frames

1.4 Survey Terminology

- Example: I want to infer the proportion of Hong Kong high school students who support the use of ChatGPT for homework
- I decide I will ask 200 students
- I have a list of all the high schools in HK. I select 25 of these
- For each of the 25, I collect a name list from the school and I select 8 to interview
- What are the **element**, **population**, **sampling units**, **frame(s)** and **sample** in this example?